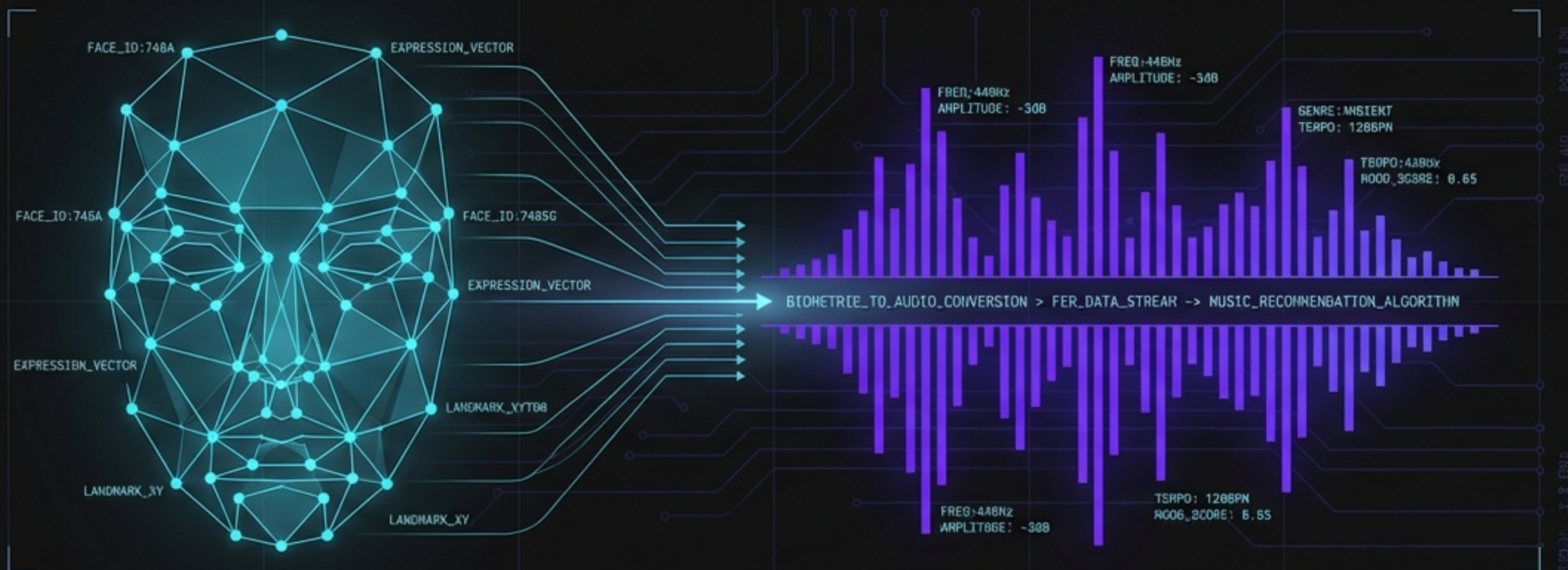




MoodMate

A Biometric-Audio Interface for Emotion-Based Recommendation.

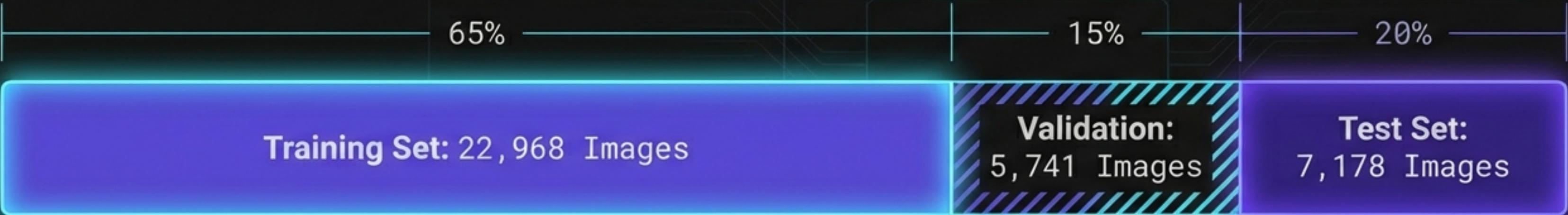
The MoodMate Objective

An intelligent system designed to bridge the gap between human emotion and digital entertainment. By leveraging Convolutional Neural Networks (CNNs), MoodMate detects emotional states from facial expressions to curate music playlists that align with the user's psychological state.



The Dataset: FER-2013

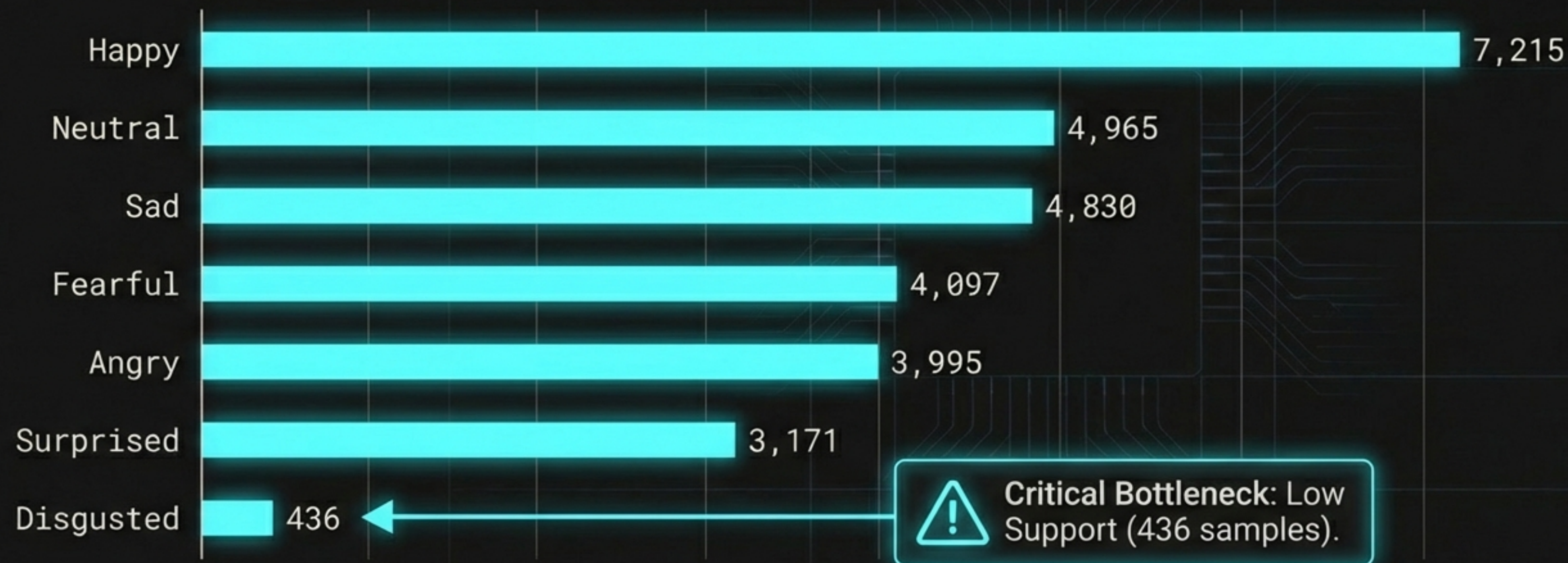
Source Material & Partitioning



Classes: Angry | Disgusted | Fearful | Happy | Neutral | Sad | Surprised

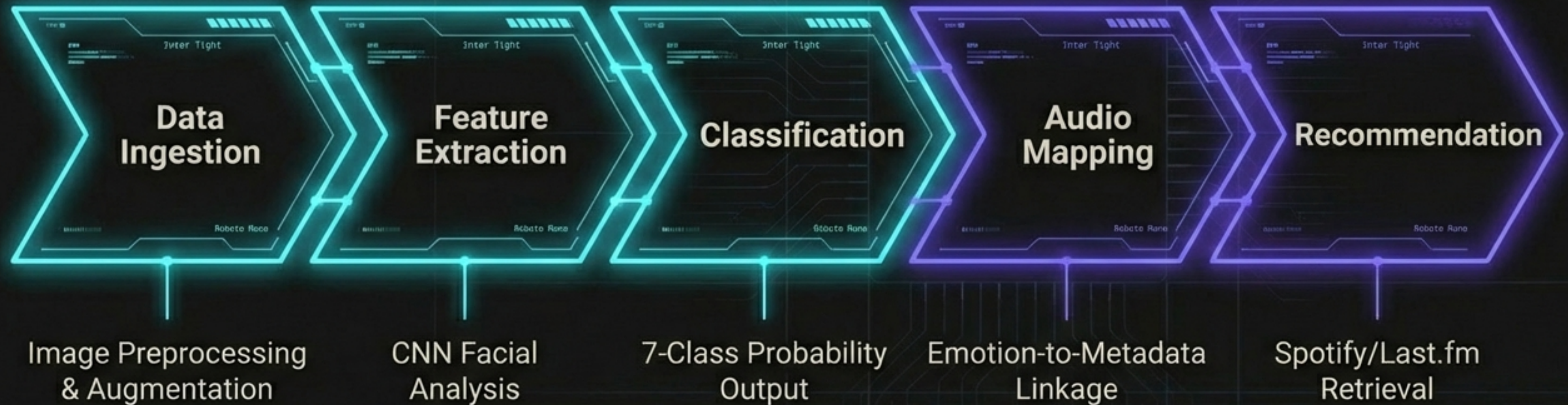
Exploratory Data Analysis: The Class Imbalance

Visualizing the 'Positivity Bias' in FER-2013



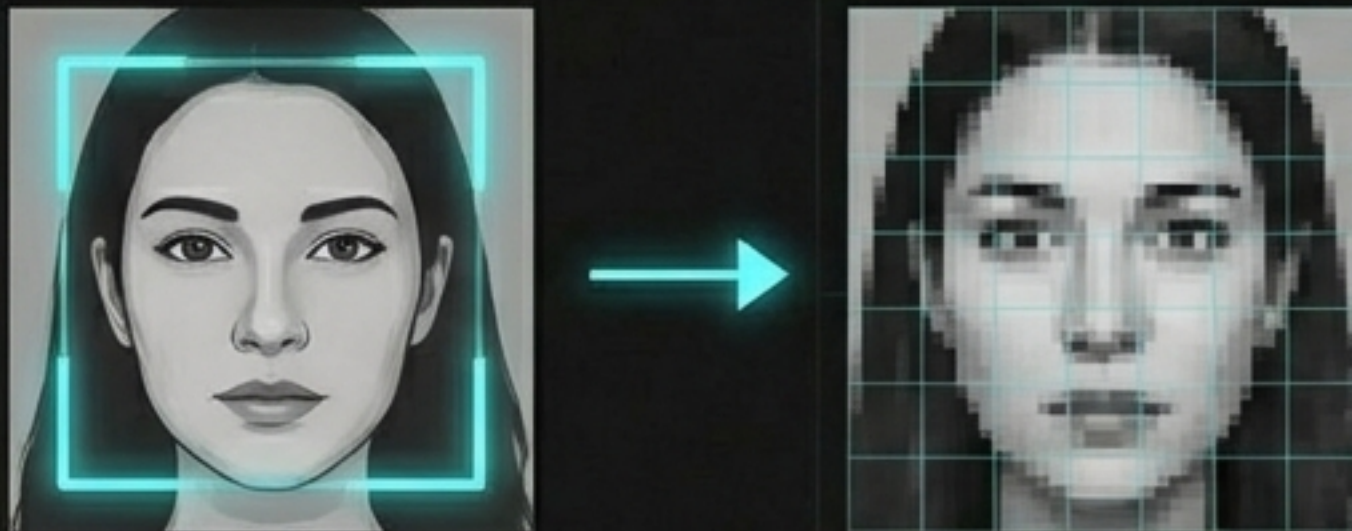
System Architecture & Workflow

Source Material & Partitioning



Preprocessing & Feature Extraction

Visual Pipeline (Input: Face)



- Grayscale Conversion (1 Channel)
- Resizing to 48x48 pixels
- Data Augmentation (Rotation, Zoom, Flip)

Audio Pipeline (Input: Metadata)

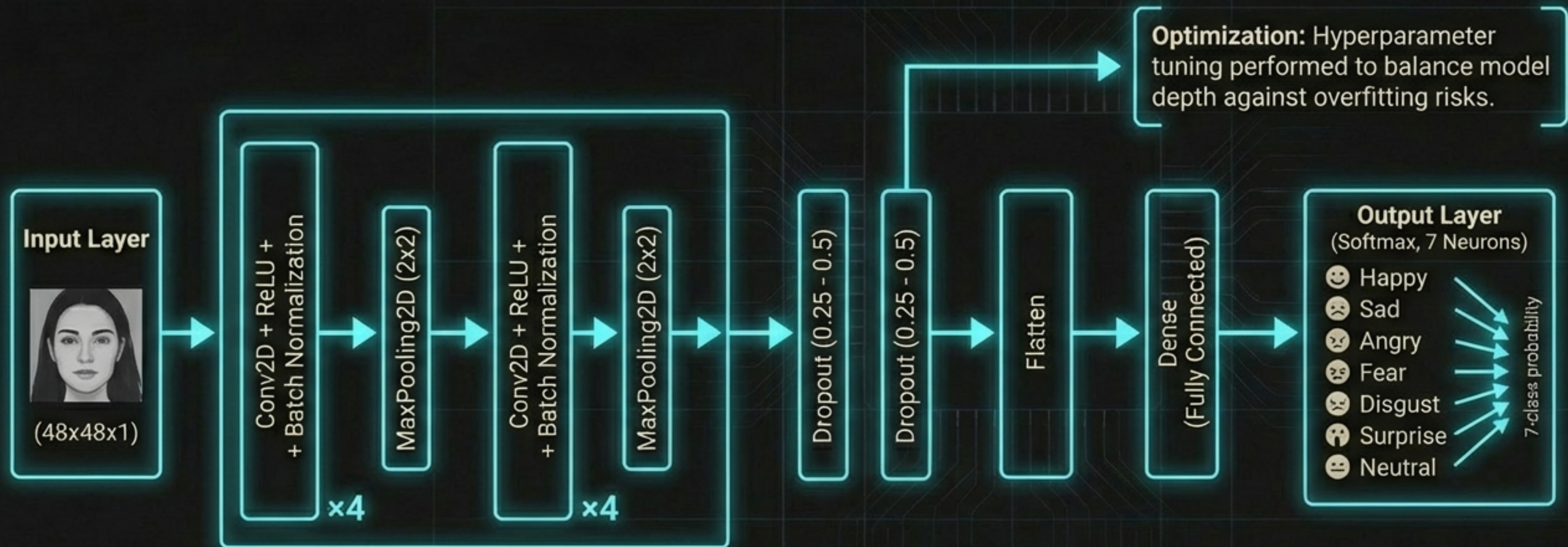


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011101 01010 010101
011010 01001 001010
101000 01001 010100
000010 11100 101010
          01010 010000
          01010 001011
          010010
```

- Source: Last.fm & Spotify Datasets
- Extracted Features: Tempo (BPM), Energy, Valence
- Tag Processing: Genre & Mood Labels

Custom CNN Architecture

Designed from Scratch for FER-2013

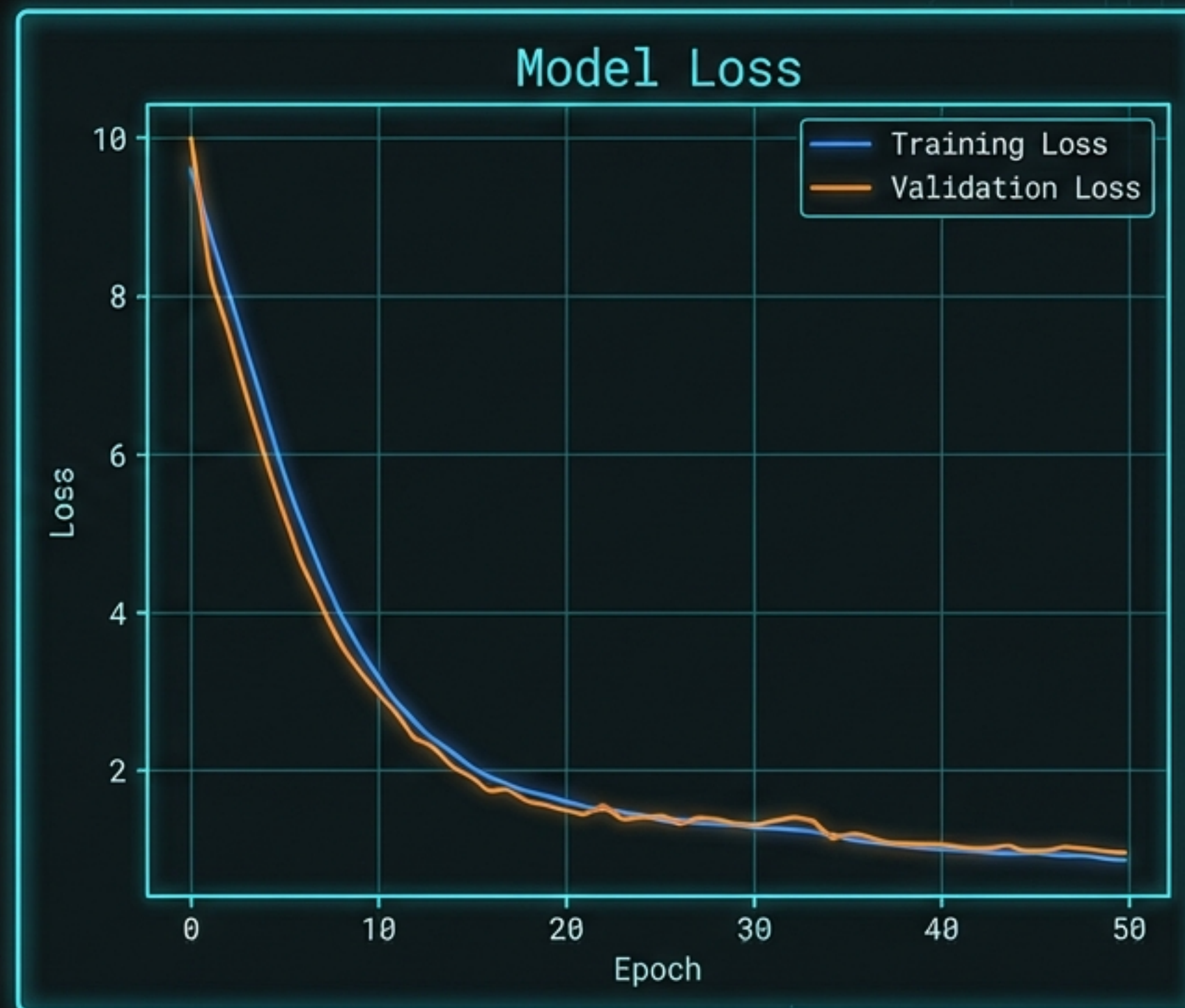
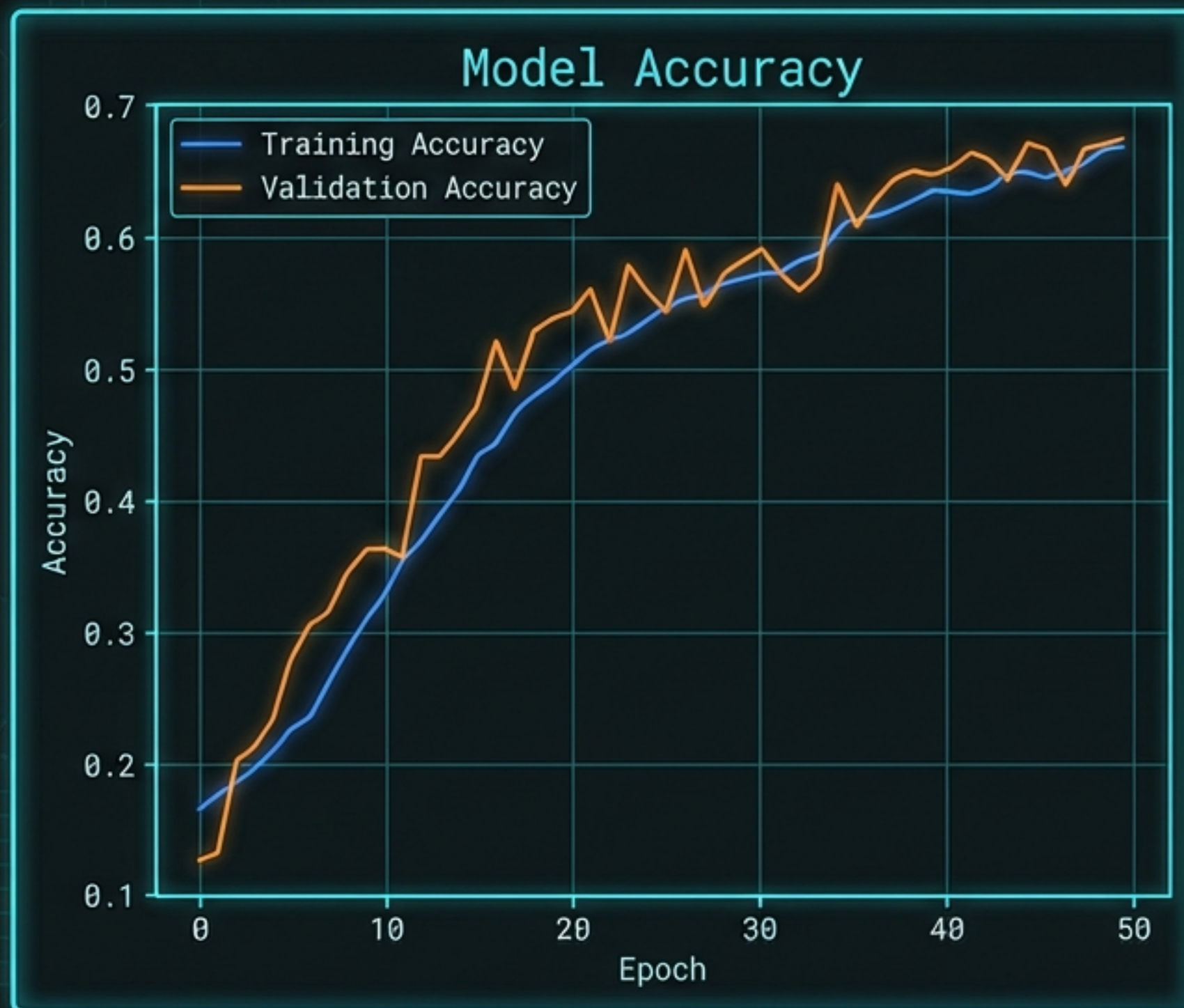


The Emotion-Music Mapping Mechanism



Recommendation Engine ranks songs by calculating the Euclidean distance between the target mood vector and song metadata.

Training Dynamics: Accuracy & Loss



Quantitative Evaluation Metrics

Class	Precision	Recall	F1-Score	Support
Happy	0.87	0.81	0.84	1774
Surprised	0.71	0.81	0.76	831
Neutral	0.55	0.66	0.60	1233
Angry	0.51	0.62	0.56	958
Sad	0.51	0.50	0.50	1247
Disgusted	0.37	0.71	0.48	111
Fearful	0.56	0.28	0.37	1024
Total Accuracy: 63%		Weighted Avg F1: 0.62		Test Samples: 7178

Error Analysis & Confusion Matrix



Analysis:

- Strong diagonal** indicates model learning.
- Semantic overlap** observed between Fear/Surprise and Sad/Neutral.
- Class imbalance** limits 'Disgusted' detection accuracy.

Real-Time User Interface



Implementation Challenges

Data Imbalance



Severe underrepresentation of 'Disgusted' (436 images) vs 'Happy' (7215 images) led to initial bias. Required class weighting during training.

Subjective Ambiguity



Human annotators often confuse 'Sadness' with 'Neutrality', creating noisy labels that limit the model's maximum theoretical accuracy.

Environmental Noise



Real-time webcam inference is highly sensitive to poor lighting and low-resolution inputs.

Future Scope & Evolution



Multimodal Input

Integrating Voice Tone Analysis and Text Sentiment to corroborate facial data for higher confidence predictions.



Data Expansion

Training on larger, cleaner datasets (e.g., AffectNet) to improve performance on minority classes like 'Fear' and 'Disgust'.



Personalization Loop

Implementing Reinforcement Learning (RL) to adjust recommendations based on user 'Skip' or 'Like' actions.

Conclusion

MoodMate
Successfully
Implemented.

- Engineered a custom CNN achieving 63% accuracy on the challenging FER-2013 dataset.
- Validated the correlation between facial biometrics and audio features.
- Demonstrated the potential for empathetic AI in consumer entertainment.